SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Assignment 3

Unit-3: Routing

Instructions: Answer all questions. Draw neat diagrams wherever necessary. Show complete working for numerical questions.

Q1. Explain the process of forwarding IP packets through a router, including the role of the routing table.

Q2. Differentiate between Static Routing and Default Routing, and explain a scenario where each would be preferred.

Q3. Explain Unicast Routing and describe the working of the Distance Vector Routing algorithm with an example network.

Q4. Explain the Link State Routing algorithm. Compare it with Distance Vector Routing on at least four parameters.

Q5. Explain Path Vector Routing and describe how it helps avoid routing loops in inter-domain routing.

Q6. Explain the basic structure of IPv6 addressing, including address representation and types of IPv6 addresses.

Q7. Explain the basics of multicasting, including how a multicast address differs from a unicast address.

Q8. Compare RIP version 1 and RIP version 2, highlighting the key improvements introduced in RIP V2.

Q9. Explain the working of OSPF (Open Shortest Path First), including the concept of areas and the Dijkstra algorithm used.

Q10. Compare BGP (Border Gateway Protocol) and EIGRP (Enhanced Interior Gateway Routing Protocol) in terms of type, metric and use case.

--- End of Assignment ---